

SHUBAYR



Online Multi-Section Store — Web & Mobile Application

Software Engineering Analysis & System Design

وثيقة التحليل والتصميم الهندسي للنظام

Project / المشروع	Shubayr Online Store (Web + Mobile)
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Prepared for / أعدت لـ	Shubayr Development Team
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Confidential — prepared as an engineering blueprint for the Shubayr project team.

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1. Executive Summary

الملخص التنفيذي

Shubayr is a full-scale online store built around the concept of a digital shopping mall: a single, trusted business offering many departments under one roof — phones and accessories, laptops and computers, home appliances, clothing, cosmetics, sports equipment, electricals, and a dedicated nursing and medical-supplies section, with room to add more departments over time.

This document is the engineering blueprint for delivering Shubayr as two coordinated products sharing one backend: a mobile application for iOS and Android built with Flutter, and a web storefront. It defines the recommended technology stack, the full set of UML diagrams (use case, class, entity-relationship, sequence, activity, state, component, and deployment), the data model, the API surface, security and localization requirements, a phased delivery roadmap, and a cost breakdown aligned with the project's US \$10,000 budget.

For the first release, orders are settled by Cash on Delivery (COD). The architecture isolates payment behind a pluggable payment module, so online gateways (for example ZainCash, FastPay, Qi Card, or card processors) can be added later without reworking the order flow.

الملخص

«شُبَيْر» متجر إلكتروني متكامل يعمل كمجمّع تسوّق رقمي يضم أقسامًا متعددة (تليفونات وإكسسوارات، لابتوبات، أجهزة منزلية، ملابس، كوزماتك، أجهزة رياضية، كهربائيات، وقسم خاص بالتمريض والمستلزمات الطبية) مع إمكانية إضافة أقسام جديدة لاحقًا. توضّح هذه الوثيقة المكدّس التقني الموصى به، وجميع مخططات UML، ونموذج البيانات، وواجهات الـ API، ومتطلبات الأمان، وخطة التنفيذ، وتوزيع الميزانية ضمن حدود 10,000 دولار. الدفع في الإصدار الأول عبر الدفع عند الاستلام، مع تصميم يسمح بإضافة بوابات دفع إلكترونية لاحقًا.

2. Project Overview & Vision

نظرة عامة على المشروع والرؤية

2.1 Vision

Deliver the smoothest way for customers in the local market to shop across many product categories from one branded store, on the phone in their pocket or in a browser, with a checkout that is trusted (Cash on Delivery), fast, and fully Arabic-first while remaining bilingual.

2.2 Business Objectives

- **Unified storefront:** Give one business a single online presence covering all of its departments.
- **Wider reach:** Increase reach and sales beyond the physical shop through mobile and web.
- **Operational control:** Give staff a simple dashboard to manage products, stock, orders, and delivery.
- **Future-proof:** Build on a foundation that scales to more departments and online payments later.

2.3 Store Departments (Sections)

The store is organized into departments, each acting like a section of a mall. Categories are hierarchical, so each department can hold sub-categories (e.g. Phones → Smartphones, Chargers, Cases).

#	Department (English)	القسم (بالعربية)	Example sub-categories
1	Phones & Accessories	تليفونات وإكسسوارات	Smartphones, chargers, cases, earphones
2	Laptops & Computers	لابتوبات وحواسيب	Laptops, monitors, keyboards, storage
3	Home & Appliances	أجهزة منزلية	Kitchen, cleaning, small appliances
4	Clothing & Fashion	ملابس وأزياء	Men, women, kids, footwear
5	Cosmetics & Beauty	كوزماتك ومستحضرات تجميل	Skincare, makeup, fragrance
6	Sports Equipment	أجهزة رياضية	Fitness gear, supplements, apparel
7	Electricals & Electronics	كهربائيات وإلكترونيات	Cables, lighting, small electronics
8	Nursing & Medical Supplies	قسم التمريض والمستلزمات الطبية	First aid, care supplies, devices

Additional departments can be added by staff at any time without code changes, thanks to the dynamic category model.

3. Project Scope

نطاق المشروع

3.1 In Scope

- Customer mobile app (Flutter, iOS + Android) and responsive web storefront.
- Staff admin dashboard for catalog, inventory, orders, delivery, promotions, and reports.
- OTP-based phone registration and login, customer profiles and multiple delivery addresses.
- Product catalog with hierarchical departments, variants, images, search, filtering, and reviews.
- Cart, wishlist, coupons, and Cash-on-Delivery checkout.
- Order lifecycle management, delivery assignment, and status tracking with push/SMS notifications.
- Arabic + English localization with full right-to-left (RTL) support; currency in IQD.

3.2 Out of Scope (Phase 1)

- Online/card payment gateways (designed for, but integrated in a later phase).
- Multi-vendor marketplace (multiple independent sellers) — single-owner model for now.
- Advanced AI recommendations and loyalty/points program (future enhancement).
- In-app live chat / customer-service ticketing (future enhancement).

3.3 Assumptions

- One business owns and fulfils all inventory; there is one central stock.
- Delivery is handled by the store's own agents within defined delivery zones.
- The client provides product data, images, branding, and an SMS/OTP provider account.
- Cash on Delivery is the only payment method at launch.

الملخص

يشمل النطاق تطبيق الهاتف (Flutter) ومتجر الويب ولوحة تحكم الموظفين، مع تسجيل الدخول عبر رمز التحقق OTP، وكتالوج منتجات بأقسام متعددة، وسيلة ومفضلة وكوبونات، والدفع عند الاستلام، وتتبع الطلبات، ودعم كامل للغة العربية (RTL) لا يشمل الإصدار الأول بوابات الدفع الإلكتروني ولا نموذج البائعين المتعددين، وهي مؤجلة لمراحل لاحقة مع تصميم يسمح بإضافتها.

4. Stakeholders & User Roles

الأطراف المعنية وأدوار المستخدمين

The system uses Role-Based Access Control (RBAC). Each role sees only the features it is permitted to use.

Role	الدور	Key capabilities
Guest	زائر	Browse and search products, view details, register/login.
Customer	زبون	Cart, wishlist, place COD orders, track orders, reviews, manage profile & addresses.
Delivery Agent	مندوب توصيل	View assigned deliveries, update delivery status.
Store Manager	مدير المتجر	Manage products, categories, inventory, orders, promotions.
Admin	مسؤول النظام	All manager rights plus users & roles, settings, reports & analytics.

5. Functional Requirements

المتطلبات الوظيفية

Requirements are grouped by module. Each is a capability the delivered system must provide.

ID	Module	Requirement
FR-1	Accounts & Auth	Phone + OTP registration/login, JWT sessions, profile, multiple addresses, RBAC.
FR-2	Catalog	Hierarchical departments/categories, products, variants (size/color), images, price & sale price.
FR-3	Search & Browse	Full-text search, filter by department/price/rating, sorting, pagination.
FR-4	Product & Reviews	Product detail pages, stock/availability, customer ratings and moderated reviews.
FR-5	Cart & Wishlist	Add/update/remove items, persistent cart, wishlist, live totals.
FR-6	Checkout & Orders	Address selection, coupon application, COD order placement, order summary.
FR-7	Order Management	Status workflow, order history, cancellation, re-order, invoices.
FR-8	Delivery	Assign delivery agent, update delivery status, delivery zones & fees.
FR-9	Promotions	Coupons (percentage/fixed), validity windows, usage limits, banners.
FR-10	Admin Dashboard	CRUD for catalog, inventory adjustments, order processing, user management.
FR-11	Inventory	Stock tracking, low-stock alerts, automatic decrement on order.
FR-12	Notifications	Push (FCM) and SMS for OTP, order confirmation, and status changes.
FR-13	Reports	Sales, top products, revenue by department, low-stock and order reports.
FR-14	Localization	Arabic/English UI, RTL layout, IQD currency formatting.

6. Non-Functional Requirements

المتطلبات غير الوظيفية

Attribute	Target / Approach
Performance	API p95 response < 300 ms; product listing loads < 2 s on 4G; image lazy-loading & CDN.
Scalability	Stateless API behind a load balancer; horizontal scaling; Redis caching of hot catalog data.
Availability	Target 99.5% uptime; automated daily database backups; health checks & auto-restart.
Security	TLS everywhere, hashed OTP, RBAC, rate limiting, input validation, OWASP Top-10 hardening.
Usability	Clean Arabic-first UX, RTL and LTR, large tap targets, 3-tap path to checkout.
Localization	Arabic + English content, RTL layout mirroring, IQD currency, local date formats.
Compatibility	Android 8.0+, iOS 13+, latest Chrome/Safari/Edge/Firefox.
Maintainability	Modular service layout, coding standards, automated tests, API documentation.
Observability	Centralized logging, error tracking (e.g. Sentry), uptime & performance monitoring.
Privacy	Minimal PII, secure storage of phone numbers/addresses, data-access controls.

الملخص

تشمل المتطلبات غير الوظيفية الأداء (زمن استجابة منخفض وتحميل سريع)، وقابلية التوسع الأفقي، والتوفر بنسبة 99.5%، والأمان الشامل (تشفير، صلاحيات، حماية OWASP)، وسهولة الاستخدام، ودعم العربية RTL وعملة الدينار العراقي، والتوافق مع الأجهزة والمتصفحات الحديثة، وقابلية الصيانة والمراقبة وحماية الخصوصية.

7. Recommended Technology Stack

المكدّس التقني الموصى به

The stack is chosen for the best balance of quality, speed of delivery, hiring availability in the region, and cost — appropriate for a US \$10,000 single-owner store that must grow gracefully.

7.1 Stack Overview

Layer	Technology	Why it was chosen
Mobile app	Flutter (Dart)	One codebase for iOS + Android, excellent Arabic/RTL support, near-native performance, fast to build — best value for the budget.
Web storefront	Next.js (React)	Server-side rendering for SEO and fast first load — important for an online store to be found on Google.
Admin dashboard	Laravel Filament / React	Rapid, secure back-office CRUD out of the box; saves significant build time.
Backend / API	Laravel 13 (PHP 8.x)	Mature e-commerce ecosystem, quick to build, large regional talent pool, low hosting cost. Exposes a clean REST/JSON API.
Database	PostgreSQL	Reliable, powerful relational engine with strong integrity for orders, stock, and money.
Cache & Queues	Redis	Caching hot catalog data, sessions, and background job queues (notifications, emails).
Search	Meilisearch	Fast, typo-tolerant Arabic/English product search.
Media storage	S3-compatible object storage	Scalable, cheap storage for product images, served via CDN.
Push notifications	Firebase Cloud Messaging	Free, reliable push to Android and iOS.
SMS / OTP	Local SMS gateway	Phone verification and order alerts for the local market.
Auth	Laravel Sanctum / JWT	Token-based auth for mobile and web from one backend.
Infrastructure	Docker on a VPS / cloud	Reproducible deployments; low monthly cost; easy to scale up later.
CI/CD	GitHub Actions	Automated tests and deployments; higher quality, fewer manual errors.
Monitoring	Sentry + uptime monitor	Catch errors and downtime early.

7.2 Architecture Style

A modular, service-oriented monolith: one Laravel backend internally organized into clear service modules (Catalog, Cart, Orders, Inventory, Users/Auth, Payments, Promotions, Notifications, Reporting). This keeps the system simple and cheap to run now, while the clean module boundaries make it straightforward to split a module into its own service later if traffic grows.

7.3 Payment Design (COD-first, pluggable)

Checkout depends on a PaymentMethod interface, not on any specific provider. Cash on Delivery is the only implementation at launch. Adding ZainCash, FastPay, Qi Card, or a card gateway later means writing a new implementation of the same interface — the cart, order, and checkout code do not change (Strategy pattern).

الملخص

تم اختيار Flutter لتطبيق الهاتف (كود واحد لنظامي iOS وAndroid مع دعم ممتاز للعربية)، وNext.js لمتجر الويب (أداء وSEO)، وLaravel كخلفية توفّر واجهة API نظيفة مع قاعدة بيانات PostgreSQL وذاكرة تخزين Redis. البنية عبارة عن «مونوليث معياري» سهل التشغيل والتكلفة، والدفع مصمّم عبر واجهة قابلة للتوسعة تبدأ بالدفع عند الاستلام وتسمح بإضافة بوابات دفع لاحقًا.

8. System Architecture

معمارية النظام

The diagram below shows how the client apps, the API, the data stores, and external services fit together. Client apps talk only to the API; the API owns all business logic and data access; heavy or slow work (push, SMS) runs in background queue workers.

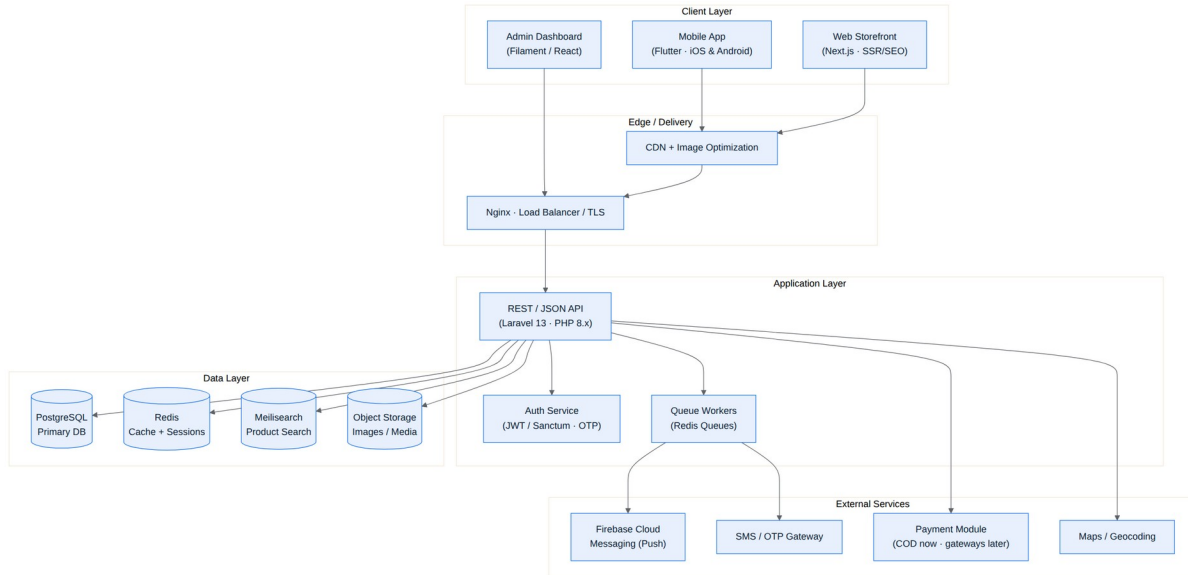


Figure 1. High-level system architecture of the Shubayr platform.

الشكل 1: المعمارية العامة لمنصة شُبَيْر.

9. UML & System Diagrams

مخططات UML ومخططات النظام

This section contains the full set of UML diagrams that the development team will implement against.

9.1 Use Case Diagram

Shows every actor (Guest, Customer, Delivery Agent, Store Manager, Admin) and the actions each can perform inside the system boundary.

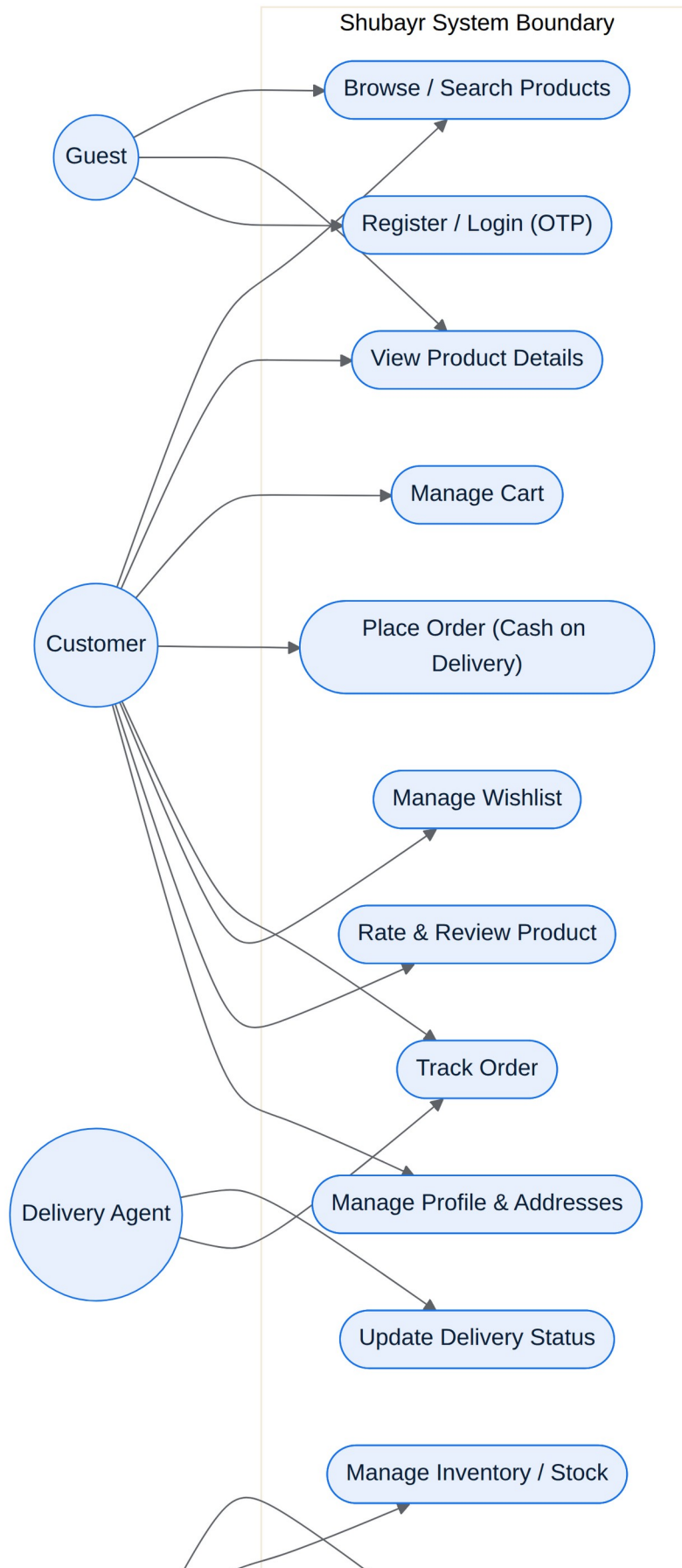


Figure 2. Use case diagram — actors and their interactions.

الشكل 2: مخطط حالات الاستخدام — الفاعلون وتفاعلاتهم مع النظام.

9.2 Class Diagram

The object model: the main domain classes, their attributes and key methods, and the relationships and multiplicities between them. This maps directly to the backend models.

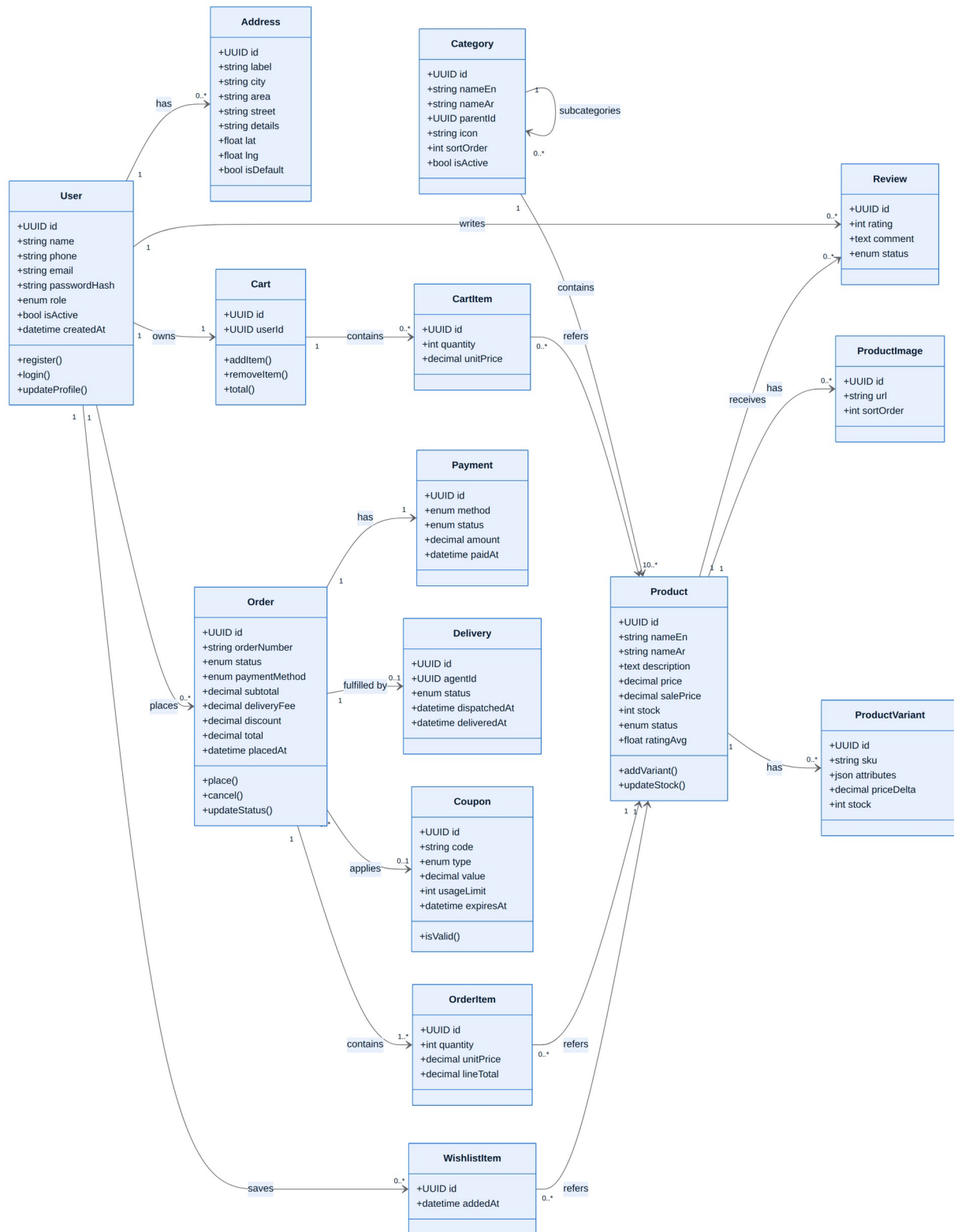


Figure 3. Class diagram — domain model and relationships.

الشكل 3: مخطط الأصناف — النموذج الكائني والعلاقات.

9.3 Entity-Relationship (Data Model)

Following the team review the data model grew significantly, so it is presented as two related views: the commerce core (users, RBAC, catalog, orders, loyalty, ratings) and the inventory & supply chain (suppliers, purchasing, batches, warehouses, stock movements, reservations, picking, returns). Together they form the PostgreSQL schema in `infra/db/schema.sql`.

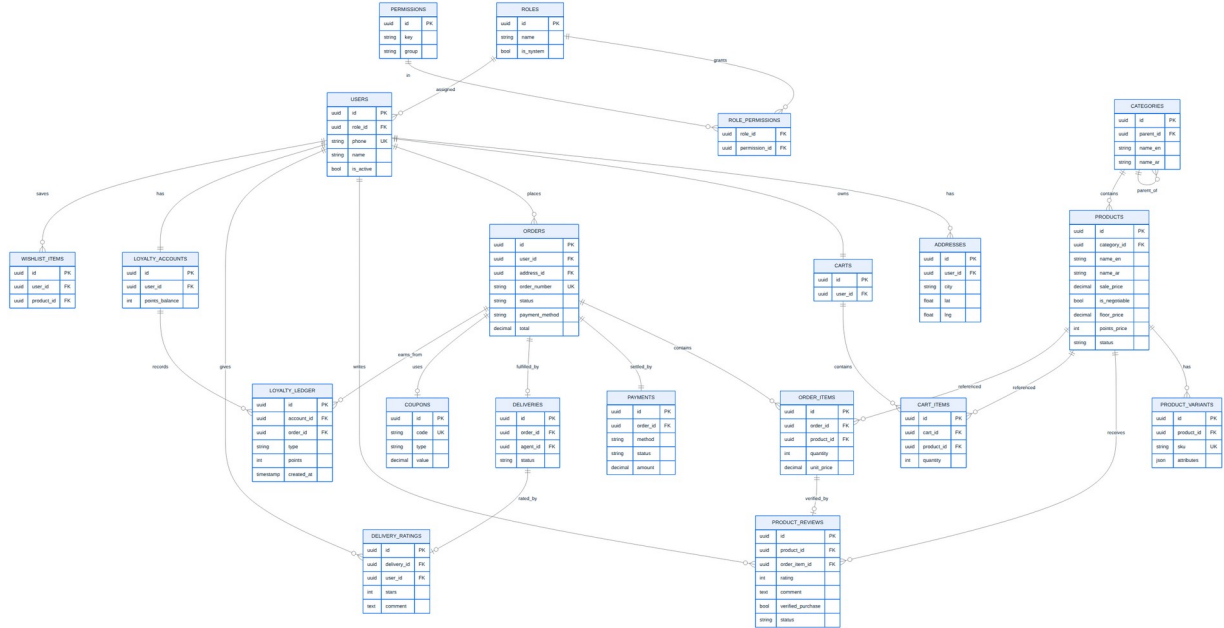


Figure 4. ER diagram (A) — commerce core: users/RBAC, catalog, orders, loyalty, ratings.

الشكل 4: مخطط الكيانات (أ) — نواة التجارة: المستخدمون والصلاحيات، الكatalog، الطلبات، الولاء، التقييمات.

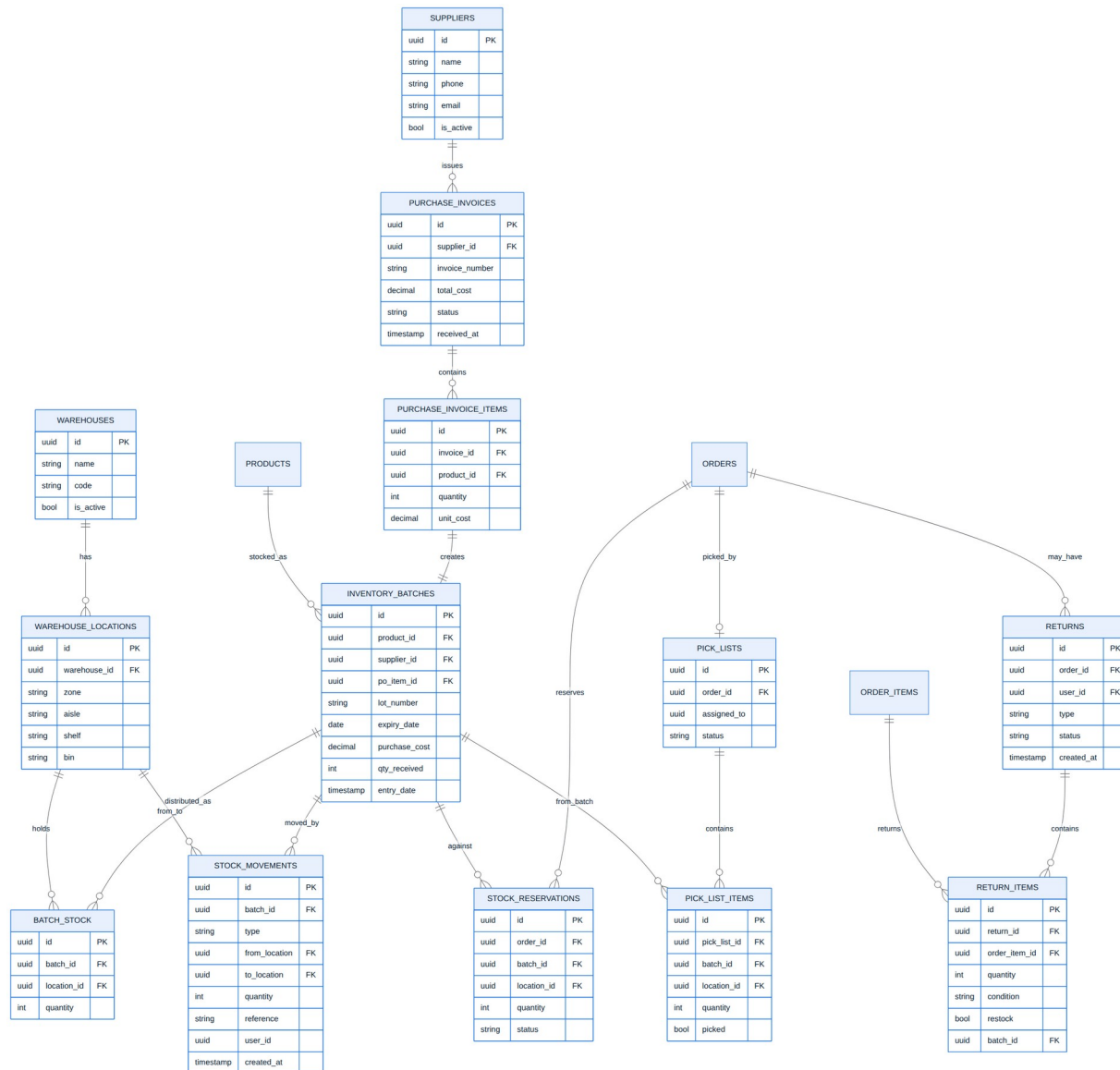


Figure 5. ER diagram (B) — inventory & supply: suppliers, purchasing, batches, locations, movements, reservations, picking, returns.

الشكل 5: مخطط الكيانات (ب) — المخزون والتجهيز: المجهزون، المشتريات، الدفعات، المواقع، الحركات، الحجز، السحب، المرتجعات.

9.4 Sequence Diagram — Checkout (Cash on Delivery)

Step-by-step message flow when a customer places a COD order, including the stock check, order creation, stock decrement, and confirmation notification.

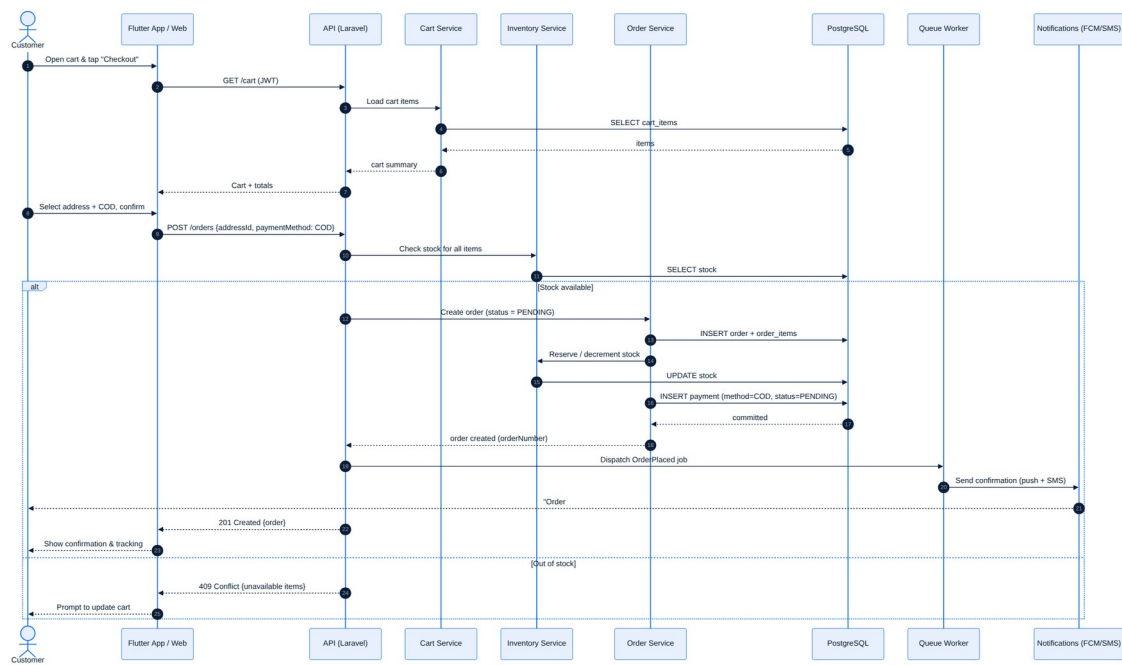


Figure 6. Sequence diagram — placing a Cash-on-Delivery order.

الشكل 6: مخطط التتابع — إتمام طلب بالدفع عند الاستلام.

9.5 Sequence Diagram — Registration / Login (OTP)

How phone-based registration and login work using a one-time password sent by SMS, ending with issued JWT tokens.

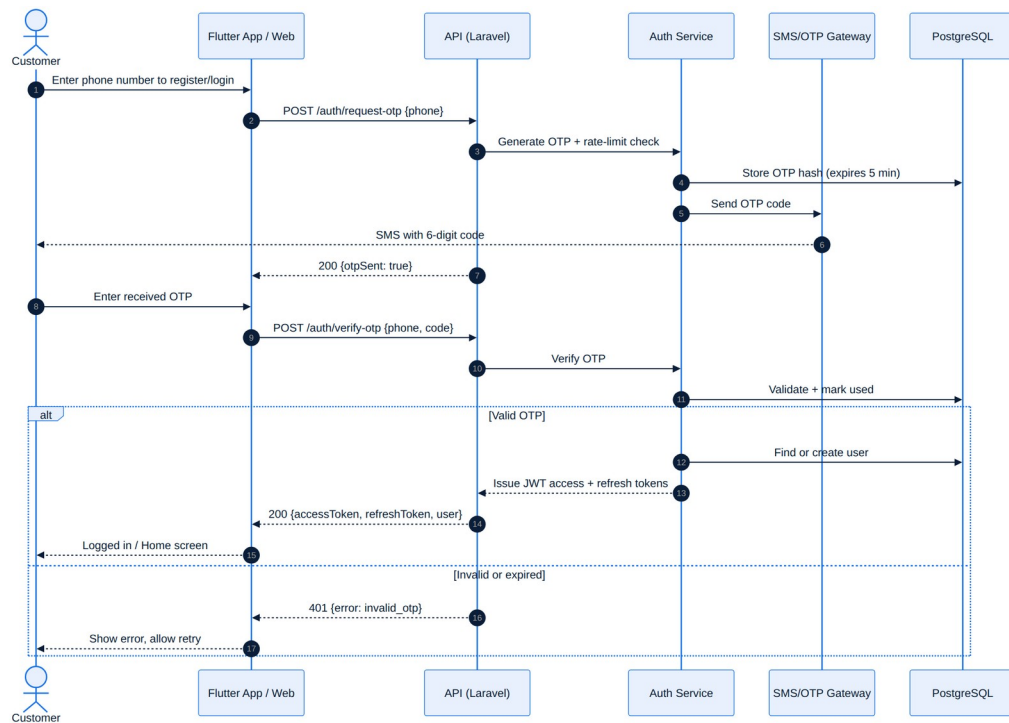


Figure 7. Sequence diagram — OTP registration and login.

الشكل 7: مخطط التتابع — التسجيل والدخول عبر رمز التحقق OTP.

9.6 Sequence Diagram — Order Tracking & Status Updates

How staff confirm and progress an order, how the delivery agent updates status, and how the customer receives live push notifications and a tracking timeline.

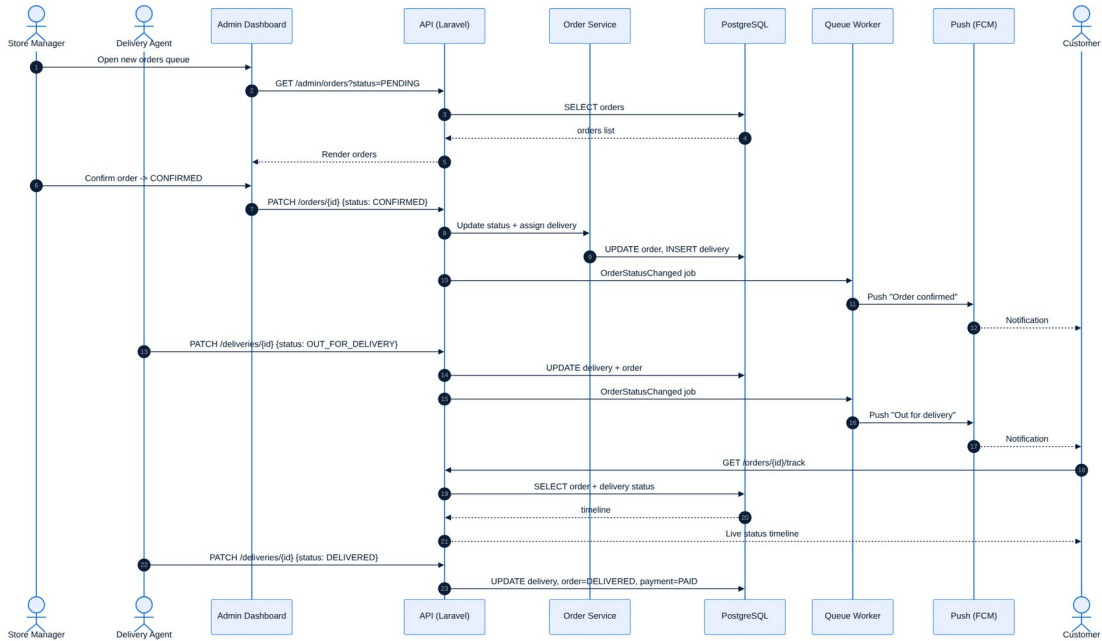


Figure 8. Sequence diagram — order processing, delivery, and tracking.

الشكل 8: مخطط التتابع — معالجة الطلب والتوصيل والتتبع.

9.7 Activity Diagram — Shopping to Order

The end-to-end customer journey as a workflow, including the decision points (login required, coupon, stock availability, confirmation).

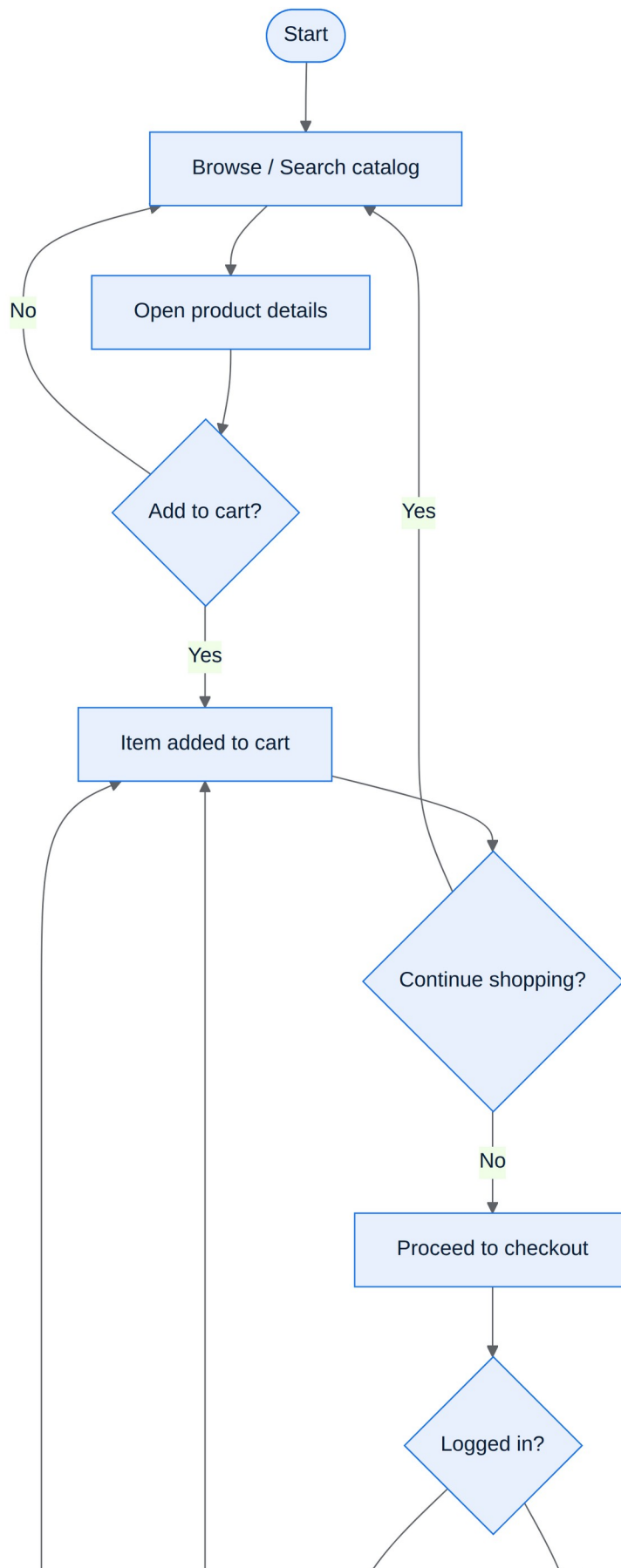


Figure 9. Activity diagram — from browsing to a placed order.

الشكل 9: مخطط النشاط — من التصفح حتى تأكيد الطلب.

9.8 State Diagram — Order Lifecycle

Every state an order can be in and the events that move it between states, from placement to delivery, cancellation, or return.

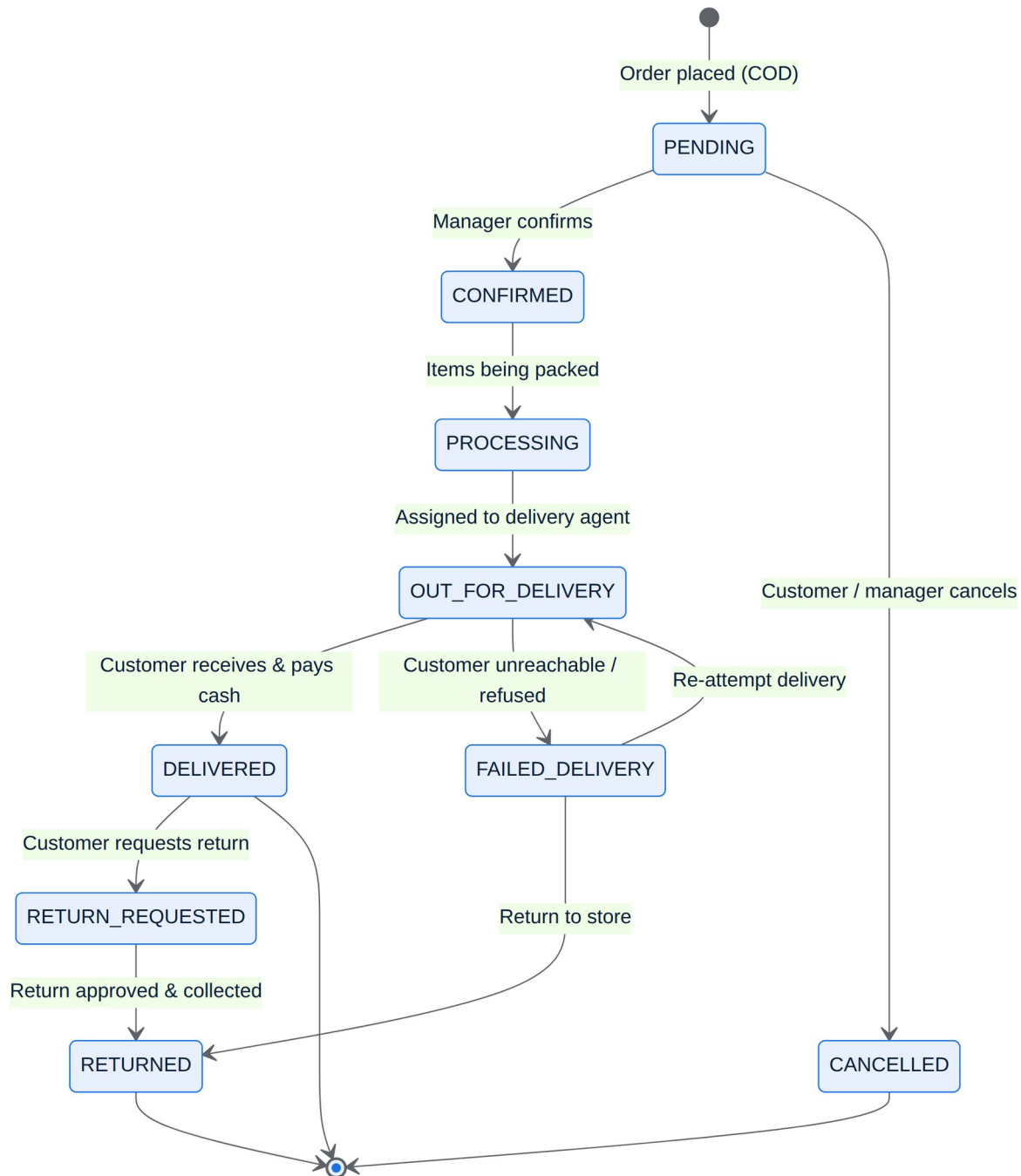


Figure 10. State machine diagram — order lifecycle.

الشكل 10: مخطط الحالات — دورة حياة الطلب.

9.9 Component Diagram

The internal software components — presentation, API gateway, domain services, and infrastructure — and how they depend on one another.

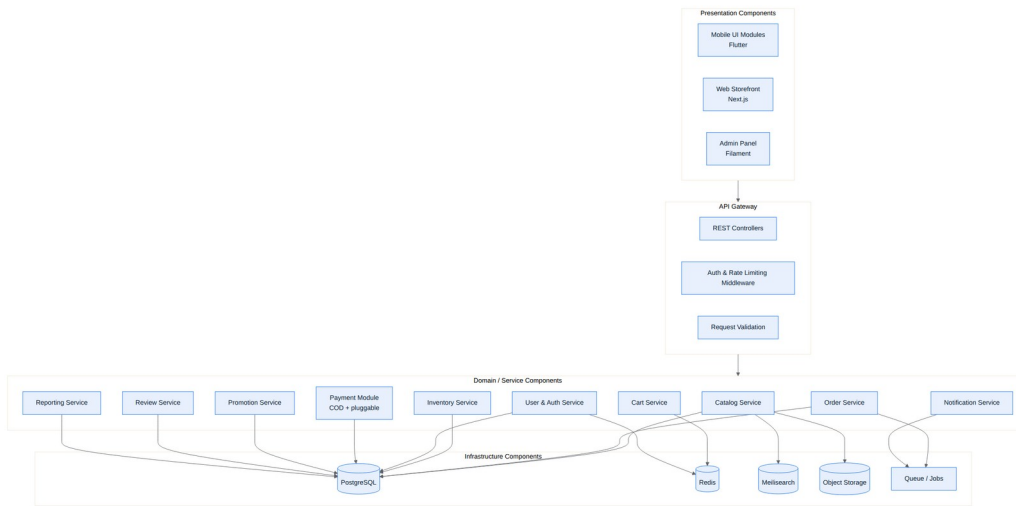


Figure 11. Component diagram — software components and dependencies.

الشكل 11: مخطط المكونات — مكونات النظام والاعتماديات بينها.

9.10 Deployment Diagram

How the software is deployed onto infrastructure: user devices, the cloud tiers (web, app, data), object storage, and third-party managed services.

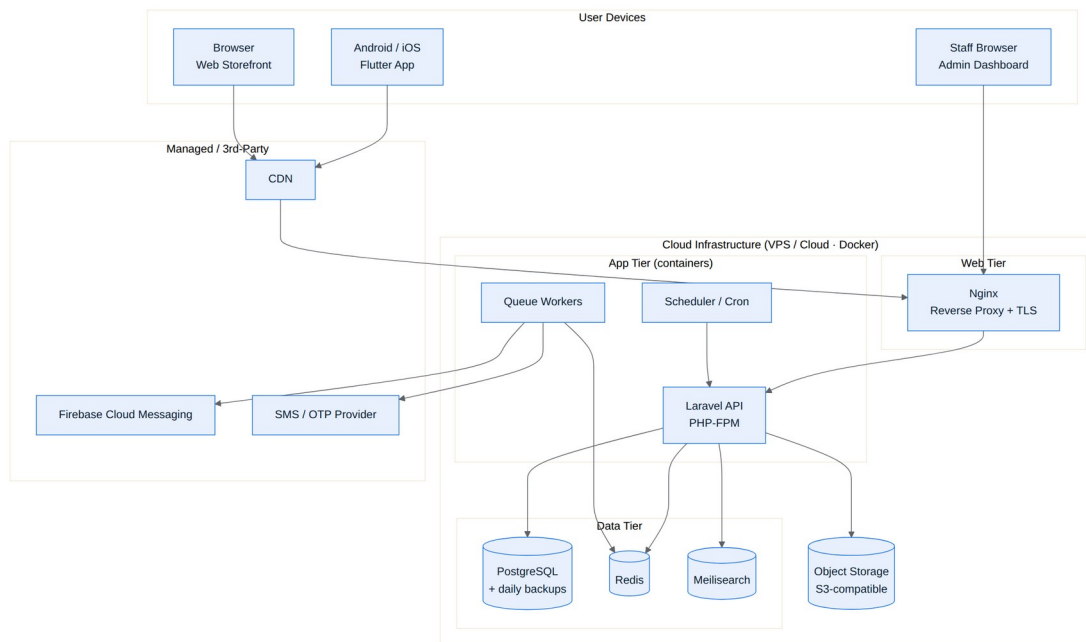


Figure 12. Deployment diagram — runtime infrastructure.

الشكل 12: مخطط النشر — البنية التحتية لوقت التشغيل.

10. Database Design

تصميم قاعدة البيانات

The PostgreSQL schema derived from the ER model. All primary keys are UUIDs; monetary values use fixed-precision decimals; every table carries created/updated timestamps.

Table	Purpose	Key fields
roles	Access roles & permissions	id, name, permissions(json)
users	Customers & staff accounts	id, role_id, name, phone(unique), email, is_active
addresses	Customer delivery addresses	id, user_id, city, area, street, lat, lng, is_default
categories	Departments & sub-categories	id, parent_id, name_en, name_ar, icon, sort_order
products	Catalog items	id, category_id, name_en, name_ar, price, sale_price, stock, status
product_variants	Size/color variants	id, product_id, sku(unique), attributes(json), stock
product_images	Product media	id, product_id, url, sort_order
carts / cart_items	Shopping carts	cart: user_id · item: product_id, quantity, unit_price
orders	Customer orders	id, user_id, address_id, order_number, status, payment_method, total
order_items	Lines within an order	id, order_id, product_id, quantity, unit_price, line_total
payments	Payment records	id, order_id, method, status, amount, paid_at
deliveries	Delivery/fulfilment	id, order_id, agent_id, status, dispatched_at, delivered_at
coupons	Discount codes	id, code(unique), type, value, usage_limit, expires_at
reviews	Product ratings & reviews	id, user_id, product_id, rating, comment, status
wishlist_items	Saved products	id, user_id, product_id, added_at

11. API Design Overview

تصميم واجهات ال API

A versioned REST/JSON API under /api/v1. Customer endpoints require a JWT; admin endpoints additionally require a staff role. A representative subset is listed below.

Method & Endpoint	Description	Access
POST /auth/request-otp	Send OTP to a phone number	Public
POST /auth/verify-otp	Verify OTP, issue tokens	Public
GET /categories	List departments & sub-categories	Public
GET /products	List/filter/search products	Public
GET /products/{id}	Product detail with variants & reviews	Public
GET /cart	Get current cart	Customer
POST /cart/items	Add item to cart	Customer
POST /coupons/validate	Validate a coupon code	Customer
POST /orders	Place a COD order	Customer
GET /orders/{id}/track	Track order status timeline	Customer
POST /products/{id}/reviews	Submit a product review	Customer
GET /admin/orders	List/manage orders	Staff
PATCH /orders/{id}	Update order status	Staff
POST /admin/products	Create/update product	Staff
PATCH /deliveries/{id}	Update delivery status	Delivery
GET /admin/reports/sales	Sales & analytics data	Admin

12. Security & Data Protection

الأمان وحماية البيانات

- **Transport security:** All traffic over HTTPS/TLS; HSTS enabled.
- **Authentication:** Phone + OTP with hashed, short-lived codes and per-phone rate limiting; JWT access/refresh tokens.
- **Authorization:** Role-Based Access Control enforced on every protected endpoint.
- **Input handling:** Server-side validation, parameterized queries (no SQL injection), output escaping.
- **Abuse protection:** Rate limiting, throttling, and lockouts on auth and order endpoints.
- **COD safeguards:** OTP-verified phone on orders, delivery confirmation, and flagging of repeat failed-delivery numbers.
- **Data protection:** Least-privilege access to PII (phone, address); encrypted backups; secrets in a vault, never in code.
- **Auditing:** Admin actions and order status changes are logged for accountability.

الملخص

يُبنى الأمان على تشفير الاتصالات (HTTPS)، والمصادقة عبر OTP مع رموز مُجزأة ومحدودة المحاولات ورموز JWT، ونظام صلاحيات RBAC، والتحقق من المدخلات ومنع حقن SQL، والحد من الإساءة، وحماية عمليات الدفع عند الاستلام عبر تأكيد الهاتف وتتبع التوصيل، مع حماية البيانات الشخصية وتسجيل إجراءات الإدارة.

13. Localization & RTL Support

التعريب ودعم الاتجاه من اليمين لليسار

Shubayr is Arabic-first and bilingual. Every catalog entity stores both Arabic and English names, and the entire UI mirrors correctly for right-to-left reading.

- Full RTL layout mirroring in both the Flutter app and the web storefront.
- Bilingual content fields (name_ar/name_en) throughout the catalog.
- Arabic numerals, IQD currency formatting, and local date/time formats.
- Language switch that persists per user; SMS and push messages localized.
- Arabic-aware search (typo tolerance and normalization) via Meilisearch.

14. Delivery Roadmap & Milestones

خارطة الطريق والمراحل

A phased plan that delivers a working MVP early and layers in the rest. Estimated total build time is roughly 4–5 months with a small team working in parallel.

Phase	Focus	Approx. duration	Key deliverables
0	Discovery & UI/UX design	2–3 weeks	Requirements sign-off, wireframes, visual design, clickable prototype
1	Backend foundation	3–4 weeks	Database, Auth (OTP), catalog & categories API, admin scaffolding
2	Cart, checkout & orders	3 weeks	Cart, coupons, COD checkout, order lifecycle, notifications
3	Admin dashboard	3 weeks	Product/inventory/order management, promotions, reports
4	Mobile app + web storefront	4 weeks	Flutter app and Next.js storefront on the finished API
5	QA, hardening & UAT	2–3 weeks	Testing, performance & security passes, user acceptance
6	Launch & handover	1–2 weeks	Store submission, deployment, documentation, training

الملخص

خطة تنفيذ على مراحل تبدأ بالتصميم ثم الخلفية والمصادقة والكتالوج، فالسلة والدفع عند الاستلام والطلبات، ثم لوحة التحكم، ثم تطبيق الهاتف ومتجر الويب، وأخيرًا الاختبار والإطلاق والتسليم. المدة التقديرية الإجمالية نحو 4–5 أشهر بفريق صغير يعمل بالتوازي.

15. Budget Breakdown (US \$10,000)

توزيع الميزانية

An indicative allocation of the US \$10,000 one-time build budget across the work streams. Actual figures depend on the final team and rates.

Work stream	Allocation (USD)	Share
Discovery & UI/UX design	\$1,200	12%
Backend API + admin dashboard	\$3,000	30%
Mobile app (Flutter)	\$2,800	28%
Web storefront (Next.js)	\$1,500	15%
QA & testing	\$700	7%
DevOps, deployment & setup	\$500	5%
Project management & documentation	\$300	3%
TOTAL	\$10,000	100%

Recurring costs are separate from the build budget and are borne by the client: hosting/VPS (~\$20-50/month), domain, SMS/OTP credits (usage-based), and app-store fees (Apple \$99/year, Google \$25 one-time).

الملخص

توزيع تقديري لميزانية البناء البالغة 10,000 دولار: التصميم \$1,200، الخلفية ولوحة التحكم \$3,000، تطبيق الهاتف \$2,800، متجر الويب \$1,500، الاختبار \$700، النشر والإعداد \$500، الإدارة والتوثيق \$300. التكاليف المتكررة (الاستضافة، النطاق، رسائل OTP، رسوم المتاجر) منفصلة وتحملها العميل.

16. Recommended Team

الفريق المقترح

Role	Responsibility
Project Manager (part-time)	Planning, coordination, client communication, delivery.
Backend Developer	Laravel API, database, admin dashboard, integrations.
Flutter Developer	iOS + Android mobile app.
Web/Frontend Developer	Next.js storefront (can overlap with backend dev).
UI/UX Designer	Wireframes, visual design, Arabic-first RTL screens.
QA Engineer (part-time)	Test plans, functional & regression testing.

17. Risks & Mitigations

المخاطر وإجراءات التخفيف

Risk	Impact	Mitigation
Scope creep (new departments/features)	Delays, over-budget	Dynamic category model; change-control; MVP-first phasing.
COD failed deliveries / fraud	Lost cost, stock issues	OTP-verified orders, delivery confirmation, flag repeat offenders.
Image-heavy pages slow	Poor UX, drop-off	CDN, image optimization, lazy loading, caching.
Small budget vs. large scope	Quality risk	Phased delivery, reuse of frameworks (Filament), shared backend.
Adding payments later	Rework risk	Pluggable payment interface designed from day one.
Key-person dependency	Continuity risk	Documentation, code standards, Git, CI/CD.

18. Future Enhancements

تحسينات مستقبلية

- Online payment gateways: ZainCash, FastPay, Qi Card, and card processing.
- Loyalty & points program and referral rewards.
- Expansion to a multi-vendor marketplace if the business grows.
- In-app live chat and customer-support tickets.
- AI-based product recommendations and personalized home screen.
- Advanced analytics dashboards and marketing automation.

19. Version 2.0 — Team Review Architecture

الإصدار 2.0 — معمارية مراجعة الفريق

This version incorporates the team's review, which upgrades Shubayr from a basic store into a store with a proper inventory, purchasing, and fulfillment backbone. The eighteen rules below are now part of the design and the database schema. They are grouped by subsystem.

الملخص

يصنف هذا الإصدار مراجعة الفريق (18 نقطة) التي ترفع «شُبَيْر» إلى نظام متكامل يشمل المخزون والمشتريات والتجهيز: هوية بيضاء قابلة لإعادة التسمية، واجهات Flutter (زبون/مندوب/إدارة) + Flutter Web للإدارة + Next.js للمتجر العام، صلاحيات تفصيلية، إدخال البضاعة عبر فواتير شراء مرتبطة بمجهز، دفعات مخزنية مستقلة، مخازن ومواقع، حركات مخزنية لكل تغيير، سحب FEFO، حجز الكمية قبل الخصم، توجيه السحب، كلفة لكل دفعة وفصل سعر البيع، مرتجعات جزئية حسب الحالة، نقاط ولاء ودعم التفاوض بالنقاط، فصل تقييم المنتج عن تقييم التوصيل، وسجل تدقيق شامل.

19.1 White-label & System Interfaces

The store is generic/white-label: its name, logo, colors, and currency live in a `store_settings` table and are loaded at runtime, so the identity can change with no code changes. The client interfaces are:

Interface	Technology	Users / Purpose
Mobile app	Flutter (iOS + Android)	Customer, Delivery Agent, and Admin/Staff — role decides the experience.
Admin dashboard	Flutter Web	Same Flutter codebase built for the web, used by staff for management.
Public storefront	Next.js (React)	SEO-friendly public website for customers.
Backend	Laravel REST API	One API serving all of the above; all business logic lives here.

19.2 RBAC — Detailed Roles & Permissions

Access control is permission-based from the start, not just Admin/Manager. Roles are assigned fine-grained permissions (e.g. `inventory.adjust`, `purchasing.manage`, `orders.confirm`) enforced on every endpoint. Seed roles include Admin, Manager, Purchasing, Warehouse, Delivery, and Customer.

19.3 Suppliers & Purchasing

Goods never enter stock by editing a quantity. They enter only through a Purchase Invoice linked to a Supplier. Receiving the invoice creates inventory batches and places them into warehouse locations, recording the landed cost.

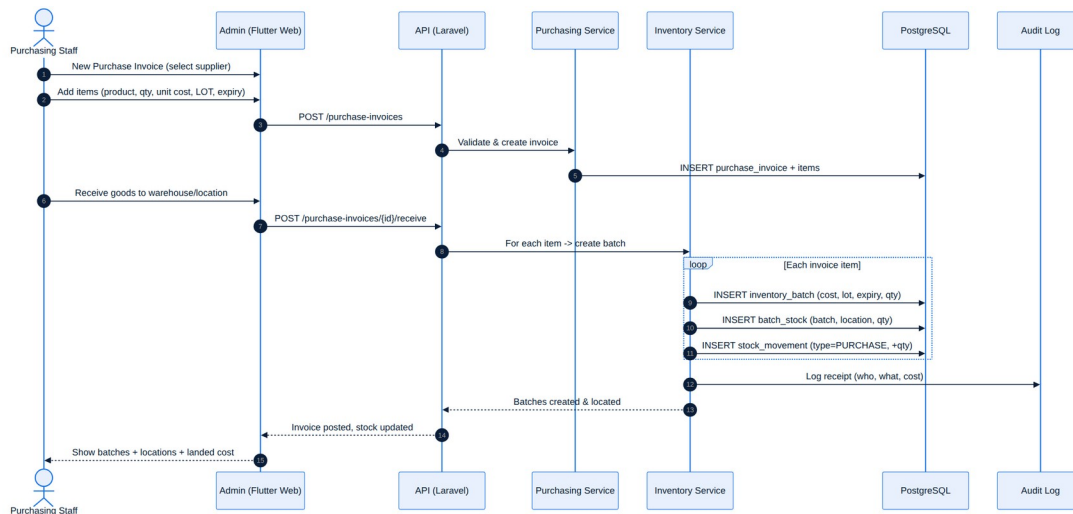


Figure 13. Sequence — receiving goods via a purchase invoice into batches & locations.

الشكل 13: مخطط التتابع — استلام البضاعة عبر فاتورة شراء إلى دفعات ومواقع.

19.4 Inventory: Batches, Warehouses & Locations

Each purchase batch of the same item stays independent, keeping its own quantity, purchase cost, entry date, LOT number, and expiry date (if any). A batch's quantity is distributed across warehouse locations structured as Warehouse → Zone / Aisle / Shelf / Bin, so the exact quantity in each location is always known.

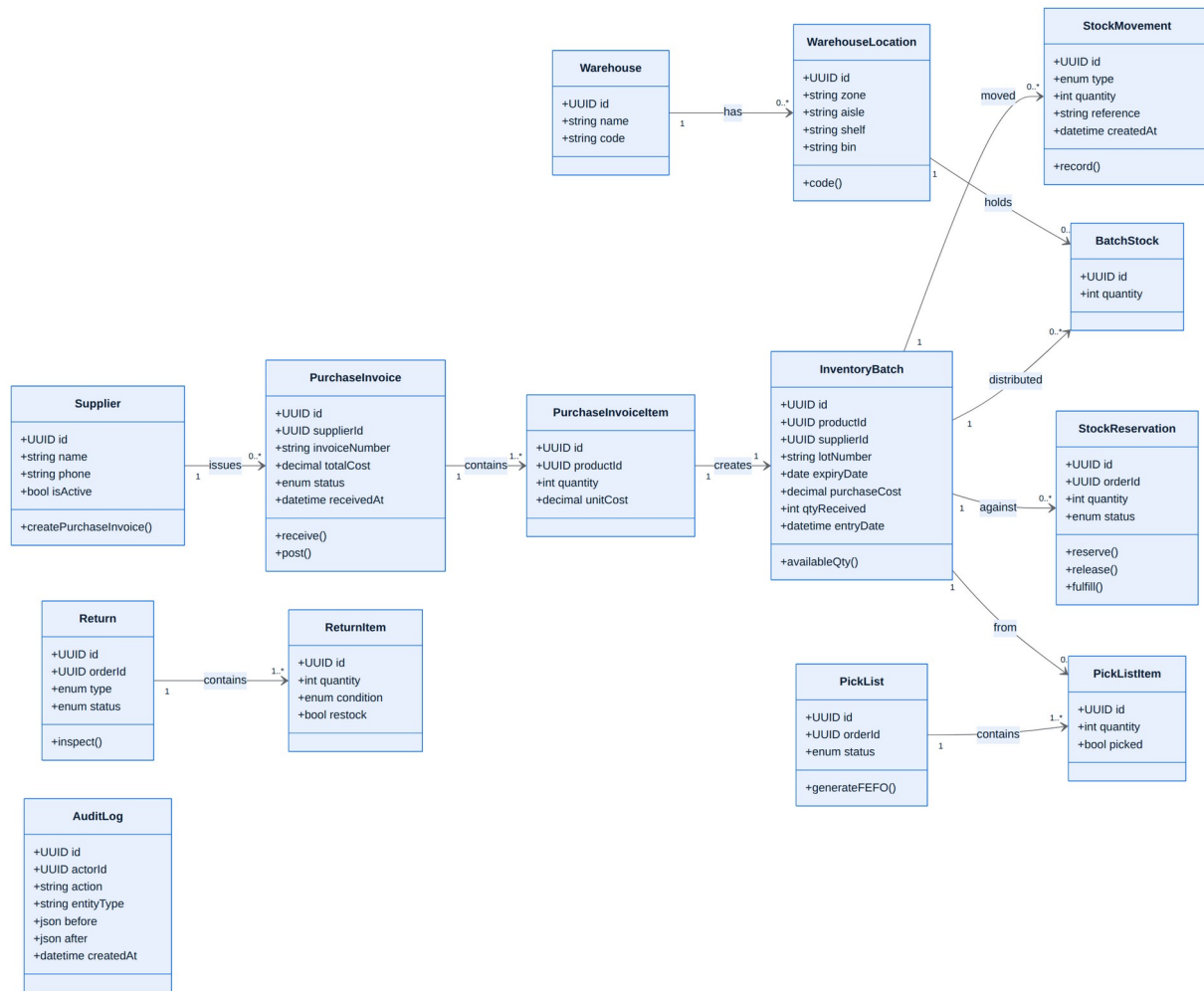


Figure 14. Class diagram — inventory & supply-chain domain.

الشكل 14: مخطط الأصناف — نطاق المخزون وسلسلة التوريد.

19.5 Stock Movements & Controlled Adjustments

Every change to stock — purchase, sale, transfer, return, reservation, release, or adjustment — is recorded as a stock movement. There is no silent editing of a balance; the only manual change is an explicit, permissioned adjustment that is itself logged.

19.6 Order Reservation, FEFO & Picking

When an order is accepted, the required quantity is reserved first (it is not deducted yet), which prevents the same stock being sold to another order. For products with an expiry date, reservation and picking follow FEFO (First-Expiry-First-Out), drawing from the nearest-expiry batch. The system then generates a pick list telling the warehouse worker exactly what to pick: item + quantity + warehouse + shelf + batch. Stock is only deducted when the order is dispatched.

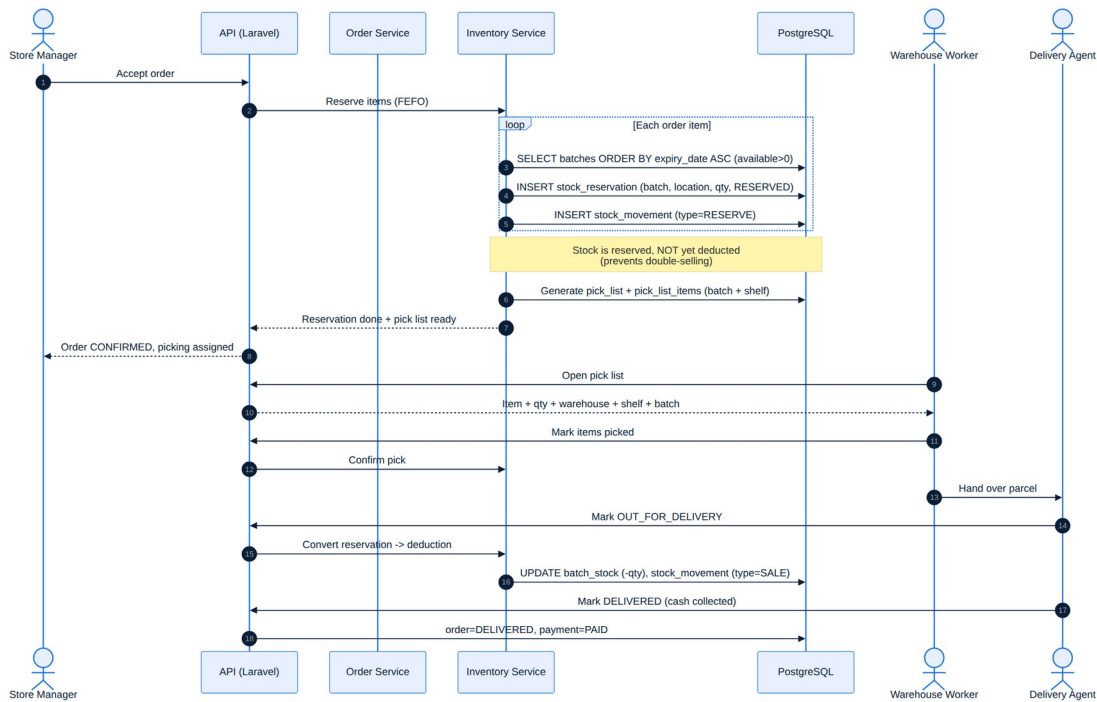


Figure 15. Sequence — reservation, FEFO batch selection, guided picking, and deduction on dispatch.

الشكل 15: مخطط التتابع — الحجز واختيار الدفعة بنظام FEFO والسحب الموجّه والخصم عند الإرسال.

19.7 Costing & Pricing

Each batch keeps its own purchase cost, and the system computes an average cost across batches. The selling price is stored separately on the product and is never changed automatically by purchase cost — pricing stays a deliberate business decision.

19.8 Returns — Partial & Condition-based

Returns are modeled from the start. A customer may return one item and keep the rest (partial returns are first-class). Each returned item is inspected and marked sellable, opened, or damaged; only sellable items are restocked (as a return-in movement into a batch). Opened items go to quarantine/open-box and damaged items are written off — not every return re-enters inventory.

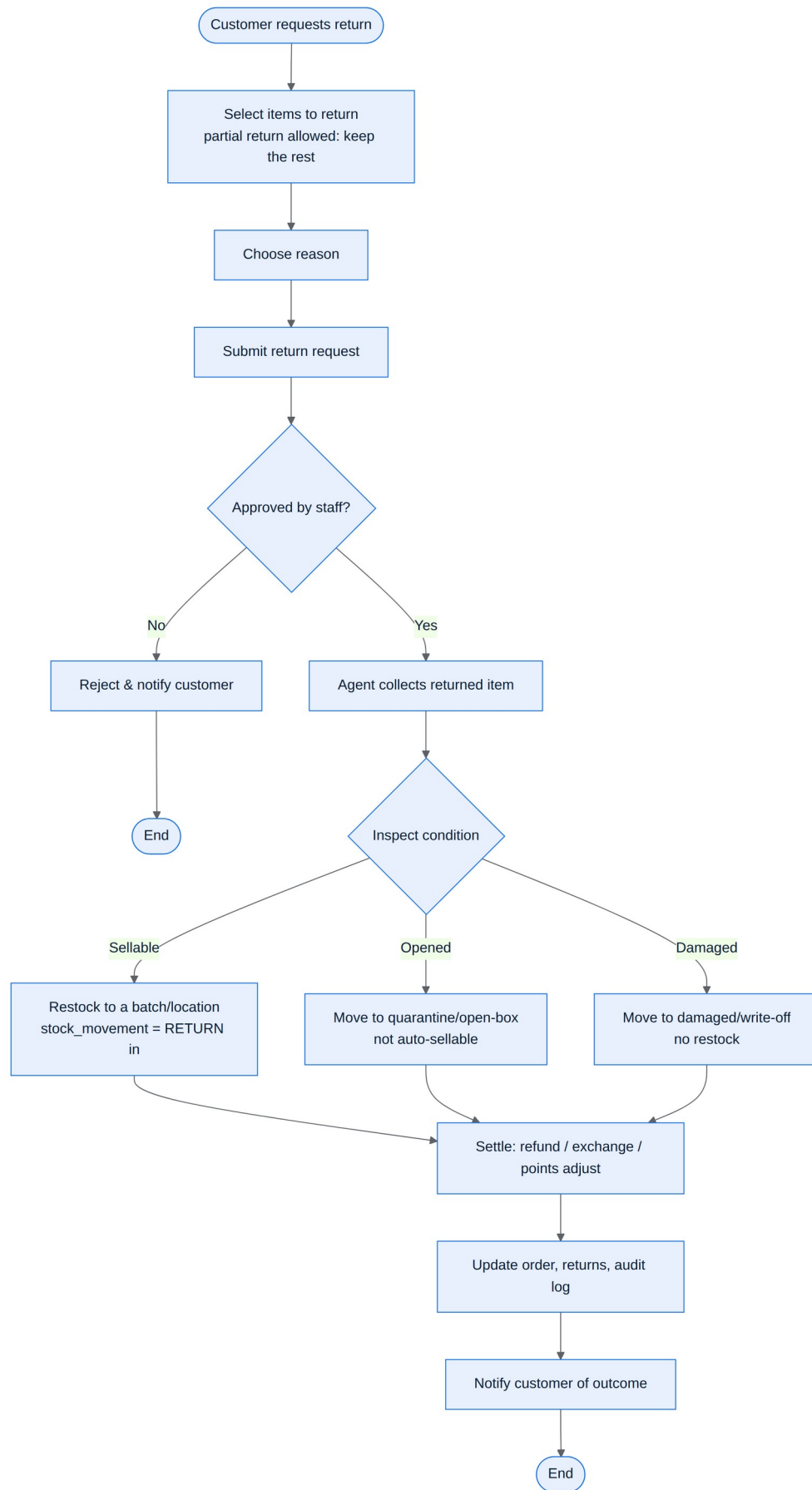


Figure 16. Activity — return handling with partial returns and condition-based restocking.

الشكل 16: مخطط النشاط — معالجة المرتجعات مع الإرجاع الجزئي وإعادة التخزين حسب الحالة.

19.9 Loyalty Points & Points Negotiation

A loyalty account and a points ledger exist from the start. Points are awarded after an order is delivered, and can be redeemed or adjusted. The data model also supports points-based negotiation now (products carry `is_negotiable`, `floor_price`, and `points_price`), so the negotiation feature can be built on top later without a schema change.

19.10 Ratings — Product vs Delivery

A product review (stars + comment) is tied to a real purchased order item and flagged as a verified purchase, which blocks fake reviews. Separately, the delivery/agent receives its own independent rating after the order is completed; the two are never mixed.

19.11 Audit Trail

Sensitive operations are logged to an audit trail with the actor, the action, the entity, and the before/after values: inventory changes, price changes, cost changes, purchases, returns, order-status changes, and permission changes. This gives full accountability.

19.12 New Database Entities (summary)

Area	New tables
RBAC	permissions, role_permissions
Purchasing	suppliers, purchase_invoices, purchase_invoice_items
Inventory	inventory_batches, warehouses, warehouse_locations, batch_stock, stock_movements
Fulfilment	stock_reservations, pick_lists, pick_list_items
Returns	returns, return_items
Loyalty	loyalty_accounts, loyalty_ledger
Ratings	product_reviews (verified), delivery_ratings
Audit / White-label	audit_logs, store_settings

The complete, validated PostgreSQL schema for all of the above ships in the project repository at `infra/db/schema.sql` (37 tables).

20. Appendix — Glossary

ملحق — مسرد المصطلحات

Term	Meaning
COD	Cash on Delivery — customer pays cash when the order is delivered.
OTP	One-Time Password — a short code sent by SMS to verify a phone number.
RBAC	Role-Based Access Control — permissions granted by user role.
JWT	JSON Web Token — a secure token used to keep a user logged in.
RTL	Right-to-Left — text/layout direction used for Arabic.
MVP	Minimum Viable Product — the first usable, launchable version.
API	Application Programming Interface — how apps talk to the backend.
CDN	Content Delivery Network — serves images fast from nearby servers.
UML	Unified Modeling Language — the standard diagrams in this document.
SKU	Stock Keeping Unit — a unique code identifying a product variant.
Batch / LOT	A specific quantity of an item received together, with its own cost and expiry.
FEFO	First-Expiry-First-Out — sell/pick the nearest-to-expire stock first.
Reservation	Holding stock for an accepted order before it is physically deducted.
Picking	Collecting the ordered items from their shelves/locations for dispatch.
Stock movement	A logged record of any change in stock (purchase, sale, transfer, return, adjust).
Purchase invoice	The document that brings supplier goods into stock as batches.
Loyalty ledger	The running record of points earned, redeemed, or adjusted per customer.
Audit trail	A log of sensitive actions (who changed what, before/after) for accountability.

White-label	Building so the store's name/branding can be changed without code changes.
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— End of document — • نهاية الوثيقة —